



Comsats University Islamabad, Lahore Campus

Assignment <1> – SPRING 2025

Course Title	Information Security	Course Code	CSC432	Credit Hours	3(2,1)
Instructor/s	Dr. Muhammad Sharjeel	Program Name	BSSE		
Semester	8 th	Batch	FA21	Section	A
Time Allowed	1 week	Maximum Marks	March 4, 2025		
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Question no. 1

- Find the IP addresses of www.google.com, facebook.com, and cuilahore.edu.pk
- Can you locate where these servers (computers) are?

```
C:\Users\Aounh>nslookup www.google.com
Server: UnKnown
Address: 192.168.0.1

Non-authoritative answer:
Name: www.google.com
Addresses: 2a00:1450:4018:803::2004
          192.178.24.164

C:\Users\Aounh>
```

```
C:\Users\Aounh>nslookup facebook.com
Server: UnKnown
Address: 192.168.0.1

Non-authoritative answer:
Name: facebook.com
Addresses: 2a03:2880:f167:81:face:b00c:0:25de
          157.240.227.35

C:\Users\Aounh>
```

```
C:\Users\Aounh>nslookup cuilahore.edu.pk
Server: UnKnown
Address: 192.168.0.1

Non-authoritative answer:
Name: cuilahore.edu.pk
Address: 203.124.44.149

C:\Users\Aounh>
```

DNS server with name “UnKnown” which is responding to my request. 192.168.0.1 is the IP Address of my local router. Non-authoritative answer means that google is not directly responding to my request, instead some intermediary server is responding to the query. It might have placed the address in its cache. I have used **nslookup** command on command prompt in Windows 11 operating system. This command is useful for querying DNS server to obtain IP address. I found where servers are located using an online website <https://ipinfo.io/account/search> to obtain geolocation by first finding the IP address using nslookup.

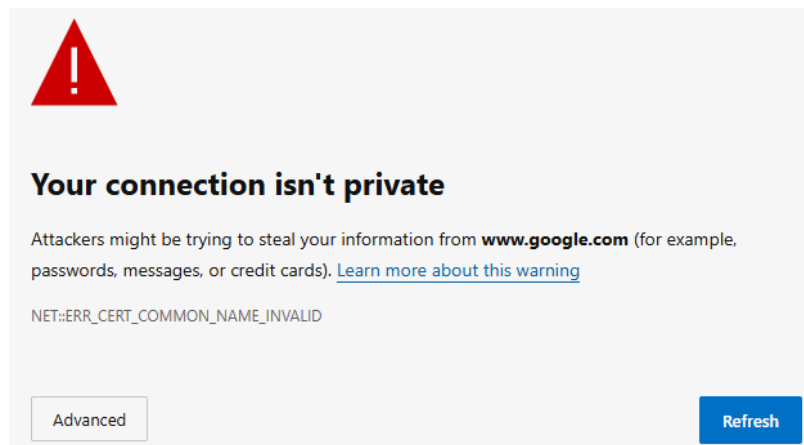
So, the IP Address of these websites is:

Website	IP Address	Geolocation
www.google.com	192.178.24.164	Hostname: mct04s02-in-f4.1e100.net
		City: Muscat
		Region: Muscat
		Country: Oman (OM)
		Location (Latitude, Longitude): 23.5841, 58.4078
facebook.com	157.240.227.35	Hostname: edge-star-mini-shv-01-mct1.facebook.com
		City: Barkā'
		Region: Al Batinah South
		Country: Oman (OM)
		Location (Latitude, Longitude): 23.6787, 57.8860
cuilahore.edu.pk	203.124.44.149	Hostname: host202003.comsatshosting.com
		City: Rawalpindi
		Region: Punjab
		Country: Pakistan (PK)
		Location (Latitude, Longitude): 33.5973, 73.0479

Question no. 2

Modify your PC's hosts file to map www.google.com to Facebook's IP address and try to do a google search. What happens? Remove the modification to the host file and repeat.

I modified host file at C:\Windows\System32\drivers\etc\ which represents domain name to IP address mapping, by entering the entry 157.240.227.35 www.google.com at the bottom and when I queried google.com in browser, it gave me this error:



This is because every server returns a certificate (which proves that it is what it claims to be), when browser compared received Facebook's server certificate with google.com, I attributed it as a threat.

When I commented the last entry in host file and again queried the browser, website opened normally.

Question no. 3

Find the domain name of 111.68.103.178 (reverse the address and add .in-addr.arpa). Why you added .in-addr.arpa to the reverse address, what is its purpose?

```
C:\Users\ounh>nslookup 111.68.103.178
Server: UnKnown
Address: 192.168.0.1

Name:      lums.edu.pk
Address: 111.68.103.178
```

I have used nslookup again to find the domain name, it internally automatically reverse the IP address and add .in-addr.arpa internally and returns the domain name which is lums.edu.pk in this case.

Question no. 4

Can you find the owner of cuilahore.edu.pk, chatgpt.com, netflix.com, and espncricinfo.com domains? How can this information be helpful to an attacker?

I have only found name server information for each domain using nslookup -type=NS cuilahore.edu.pk. This information can be used by hacker to directly target the website an perform DDOS or any other vulnerable attack.

Domain	Name Servers	IP Address
netflix.com	ns-81.awsdns-10.com	205.251.192.81
	ns-659.awsdns-18.net	205.251.194.147
espncricinfo.com	ns-475.awsdns-59.com	205.251.193.219
	ns-1329.awsdns-38.org	205.251.195.91
chatgpt.com	hassan.ns.cloudflare.com	108.162.195.149
	savanna.ns.cloudflare.com	2606:4700:58::a29f:2c95
cuilahore.edu.pk	n1.comsats.net.pk	210.56.11.130
	n2.comsats.net.pk	203.124.45.92

Question no. 5

Find the complete route from your PC to www.deepseek.com.

I have used traceroute command which provides IP addresses of maximum 30 between source and target with the time taken. It took 7 hops to reach the destination 104.18.27.90 from source 192.168.0.1

```
C:\Users\ounh>tracert www.deepseek.com

Tracing route to www.deepseek.com.cdn.cloudflare.net [104.18.27.90]
over a maximum of 30 hops:

 1    6 ms    5 ms    3 ms  192.168.0.1
 2    6 ms   10 ms   23 ms  192.168.131.194
 3   11 ms   56 ms    5 ms  192.168.131.193
 4   29 ms   26 ms   25 ms  192.168.200.109
 5   26 ms   21 ms   26 ms  192.168.200.18
 6   27 ms   22 ms   25 ms  103.213.109.251
 7   19 ms   19 ms   19 ms  104.18.27.90
```

Question no. 6

Find the MAC address of your PC. What command did you use?

The question contains ambiguity, because although mac address is only one for each pc but each computer has more than one interfaces and each interface has a separate mac address. For example, in my case, my computer has a separate interface for WIFI connections, separate interface for Bluetooth connection and separate for LAN connection. So, I am using **getmac** command which returns mac address of all interfaces except virtual then I will use **ipconfig** which is more advance because it returns more detailed information. Sample run of **getmac** is given below:

```
C:\Users\ounh>getmac

Physical Address    Transport Name
=====
B0-5A-DA-B5-9F-94   Media disconnected
64-80-99-AD-12-A9   \Device\Tcpip_{71D5DA90-6257-4CE9-9192-3C51CC72CBF8}
64-80-99-AD-12-AD   Media disconnected
```

Now as we can see, my computer has three interfaces and only one among them is connected which is WIFI adaptor.

Now running ipconfig/all command to get more detail information:

```
C:\Users\ounh>ipconfig /all

Windows IP Configuration

Host Name . . . . . : ounh-haider
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (3) I218-LM
Physical Address. . . . . : B0-5A-DA-B5-9F-94
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 64-80-99-AD-12-AA
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 66-80-99-AD-12-A9
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

This command has also showed Microsoft Virtual Adaptor's mac addresses which are used for hotspot connection, screen mirroring, direct WIFI (without router for sharing between PCs or connecting a wireless printer)

So, the active MAC address (Wireless LAN Adaptor) is **64-80-99-AD-12-A9**

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
Description . . . . . : Intel(R) Dual Band Wireless-AC 7265
Physical Address. . . . . : 64-80-99-AD-12-A9
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::d32:4e04:cedc:a7ee%10(Preferred)
IPv4 Address. . . . . : 192.168.0.201(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, March 1, 2025 2:45:00 PM
Lease Expires . . . . . : Saturday, March 1, 2025 6:21:01 PM
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 124027033
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-3C-8C-05-B0-5A-DA-B5-9F-94
DNS Servers . . . . . : 192.168.0.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 64-80-99-AD-12-AD
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\Aounh>
```

Question no. 7

Print your PC's ARP cache table. Can you change any entry in the table? What will happen if you modify your PC's ARP table?

We can use *arp -a* command to print the PC's ARP table which represents the IP address of the sites which are once visited and now if again visited, their address will be entertained by this ARP table.

```
C:\Users\ounh>arp -a
```

```
Interface: 192.168.0.201 --- 0xa
```

Internet Address	Physical Address	Type
192.168.0.1	60-83-e7-81-ae-10	dynamic
192.168.0.63	04-e8-b9-4d-c4-d9	dynamic
192.168.0.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

After adding IP address with invalid MAC address using *arp -s*, I have tested using ping command and timeout occur then I removed the added entry and again ping the same IP address and now timeout not occurred.

```

C:\Windows\System32>arp -s 192.168.0.10 00-14-22-01-23-45

C:\Windows\System32>arp -a

Interface: 192.168.0.201 --- 0xa
  Internet Address      Physical Address      Type
  192.168.0.1           60-83-e7-81-ae-10    dynamic
  192.168.0.10          00-14-22-01-23-45    static
  192.168.0.63          04-e8-b9-4d-c4-d9    dynamic
  192.168.0.255         ff-ff-ff-ff-ff-ff    static
  224.0.0.2             01-00-5e-00-00-02    static
  224.0.0.22            01-00-5e-00-00-16    static
  224.0.0.251           01-00-5e-00-00-fb    static
  224.0.0.252           01-00-5e-00-00-fc    static
  239.255.255.250       01-00-5e-7f-ff-fa    static
  255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Windows\System32>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Windows\System32>arp -d 192.168.0.10

C:\Windows\System32>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:
Reply from 192.168.0.201: Destination host unreachable.
Reply from 192.168.0.201: Destination host unreachable.
Reply from 192.168.0.201: Destination host unreachable.
Reply from 192.168.0.201: Destination host unreachable.

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Windows\System32>

```

Question no. 8

Print the routing table on your PC and explain each line (first 10 lines, if too many).

There are 3 sections in the result of *print route* command:

Section-1: Interface list shows all the interface with their corresponding mac addresses

Section-2: IPV4 route table shows all the IP addresses with their subnet mask, responsible interface from above discussed available interfaces and metric which shows the priority, lower the metric, higher will be priority. Default route is 0.0.0.0 which is handled by WIFI interface, On-link gateway represents the loopback routes.

Section-3: Similar to previous routing table but for IPV6 addresses.

All outbound traffic passes through default gateway (192.168.0.1), my router. Loopback traffic for testing internal communications. Devices in the same subnet are directly reachable and requires no gateway.

```
C:\Windows\System32>route print
=====
Interface List
17...b0 5a da b5 9f 94 .....Intel(R) Ethernet Connection (3) I218-LM
 7...64 80 99 ad 12 aa .....Microsoft Wi-Fi Direct Virtual Adapter
 9...66 80 99 ad 12 a9 .....Microsoft Wi-Fi Direct Virtual Adapter #2
10...64 80 99 ad 12 a9 .....Intel(R) Dual Band Wireless-AC 7265
16...64 80 99 ad 12 ad .....Bluetooth Device (Personal Area Network)
 1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.0.1      192.168.0.201    50
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.0.0                 255.255.255.0    On-link          192.168.0.201    306
192.168.0.201               255.255.255.255  On-link          192.168.0.201    306
192.168.0.255               255.255.255.255  On-link          192.168.0.201    306
224.0.0.0                   240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                   240.0.0.0        On-link          192.168.0.201    306
255.255.255.255             255.255.255.255  On-link          127.0.0.1        331
255.255.255.255             255.255.255.255  On-link          192.168.0.201    306
=====

Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
1      331 ::1/128                  On-link
10     306 fe80::/64                On-link
10     306 fe80::d32:4e04:cedc:a7ee/128
                                           On-link
1      331 ff00::/8                  On-link
10     306 ff00::/8                  On-link
=====

Persistent Routes:
None
```

Question no. 9

Find the location of 111.68.103.27

I have used <https://ipinfo.io/account/search> website to obtain information:

Hostname	pu.edu.pk
City	Lahore
Region	Punjab
Country	PK
Location (Latitude, Longitude)	31.5580, 74.3507

Question no. 10

Run ‘netstat -abon’ on your PC. What information does it displays? What is the name of the process listening on port 3389? What is the status of port 3389? Why it is important for an attacker?

When I executed the command, it showed me list of all the connections with the ports listening to them and the active programs or processes listening to the connections with their process IDs, connection status (ESTABLISHED, LISTENING, CLOSE_WAIT).

I used **netstat -abon | findstr :3389** to find the status of this port, but there is no program running on my PC on this port. It is for remote desktop protocol. To protect from attackers, we can:

1. use different port as compared to standard port number
2. Ensure firewall or VPN is activated and working
3. Disable RDP (Remote desktop protocol) if not needed.

```
C:\Windows\System32>netstat -abon

Active Connections

 Proto Local Address           Foreign Address         State       PID
 TCP   0.0.0.0:135             0.0.0.0:0               LISTENING   1096
 RpcEptMapper
 [svchost.exe]
 TCP   0.0.0.0:445             0.0.0.0:0               LISTENING   4
 Can not obtain ownership information
 TCP   0.0.0.0:5040            0.0.0.0:0               LISTENING   5956
 CDPSvc
 [svchost.exe]
 TCP   0.0.0.0:7680            0.0.0.0:0               LISTENING   1500
 Can not obtain ownership information
 TCP   0.0.0.0:49664           0.0.0.0:0               LISTENING   964
 [lsass.exe]
 TCP   0.0.0.0:49665           0.0.0.0:0               LISTENING   784
 Can not obtain ownership information
 TCP   0.0.0.0:49666           0.0.0.0:0               LISTENING   1988
 EventLog
 [svchost.exe]
```

Question no. 11

Run ‘netsh wlan show profile’ on your PC? What information does it show? Can you recognize all the user profiles?

This command showed all the saved WIFI profiles on my PC or connections I made in past. It is helpful for system to keep track this information so that in future if my pc is near any of these WIFI, my device will automatically connect to that WIFI (plug & play devices feature) without the hustle of re-entering password again until not changed as compared to saved one.

```

C:\Windows\System32>netsh wlan show profile

Profiles on interface Wi-Fi:

Group policy profiles (read only)
-----
    <None>

User profiles
-----
    All User Profile      : OPPO A54
    All User Profile      : A-F (17 to 20_Room )2.4GHz
    All User Profile      : Mr. Aoun
    All User Profile      : eduroam

C:\Windows\System32>_

```

In the above sample run, 4 saved profiles are shown.

Question no. 12

Run ‘query user’ on the command line and interpret what it displays?

This command is not working on my PC as I am using Windows Home, I found another variant of this command ***whoami*** which shows currently logged in user information which is *mr. aoun haider* in my case.

```

C:\Windows\System32>whoami
aoun-haider\mr. aoun haider

```